

A Spatiotemporal Graph Approach to Buoy Wave-Field Imputation, Forecasting, and Wave-Energy Estimation

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Abstract

Ocean buoy networks report wave measurements with large and structured gaps: whole stations drop offline for weeks, and individual readings fail at random. These gaps limit both the reconstruction of past sea states and the forecasting of future ones, which in turn limits the estimation of wave energy for marine renewable-energy siting. We study whether a single modelling idea, a spatiotemporal graph neural network that shares information between nearby buoys, improves all three tasks on a national archive of U.S. National Data Buoy Center (NDBC) stations. We assemble an hourly tensor of significant wave height (WVHT) and average wave period (APD) over 140 stations selected from a national coverage audit, build a k -nearest-neighbour Gaussian graph between stations, and evaluate under a leakage-safe hide-and-recover protocol. For imputation, a graph network (GRIN) wins wherever the gaps are structured: under whole-station outages it recovers WVHT at 0.346 ± 0.007 m mean absolute error (MAE), well below spatial interpolation (0.443 m), and it is also best under 50% contiguous-block missingness. When gaps are isolated (50% completely at random), simple linear interpolation is already strong and GRIN is competitive rather than dominant. For forecasting, we compare eight methods and find a clean split by paradigm: at horizons of 6 hours or more, all three spatiotemporal graph models beat all four temporal methods, including the transformer state of the art (PatchTST, iTransformer), which are no better than a per-station autoregression. The decisive factor is spatial information, not temporal architecture, and the advantage holds across the graph family (16–20% skill over persistence at 24 hours). We trace this advantage to its physical cause: the network’s wave-height cross-correlations recover the deep-water dispersion relation (observed swell lags match the group-velocity prediction d/c_g at $\rho = 0.80$), so the graph’s real edge is that it reads swell propagation. Converting forecasts to wave power, the graph model’s advantage *grows*: because power scales with the square of wave height, its energy-forecast skill reaches +0.235 versus persistence, wider than its height skill. Across all three tasks the graph helps most where spatial structure is strongest.

Keywords: spatiotemporal graph neural networks; missing-data imputation; wave forecasting; wave energy; ocean buoy networks; NDBC.

1 Introduction

Wave measurements from moored buoys are a backbone of ocean monitoring. They feed ship-routing, coastal-hazard warning, model validation, and the assessment of wave energy as a renewable resource. Yet the records are far from complete. Sensors fail, moorings are serviced, and telemetry drops, so a buoy can vanish from the record for hours to weeks; separately, individual hourly readings are lost at random. Any downstream use of the data must first confront these gaps.

Two questions follow naturally. First, can we *reconstruct* the missing values (imputation)? Second, can we *predict* future values (forecasting)? Both are usually treated as single-station time-series problems: fill or forecast each buoy from its own past. But buoys are not independent. Neighbouring stations in the same basin see correlated sea states, and swell propagates between them with a measurable lag. This spatial structure is exactly what a single-station method throws away.

A spatiotemporal graph neural network keeps that structure. It represents each buoy as a node, connects nearby buoys with edges, and learns to pass information along those edges through time. Such models have advanced traffic and sensor forecasting [11, 20], and a graph imputation model has been proposed for generic sensor networks [6]. What is not established is whether one consistent graph formulation helps across the full applied pipeline for ocean waves (reconstruction, prediction, *and* the energy estimate that a practitioner ultimately cares about), and, crucially, *where* it helps and where it does not.

We test this end to end on a national NDBC archive. Our central finding is simple: the graph helps most precisely where spatial structure matters most, and the benefit is largest for the quantity that is hardest to get right, wave power. We also report, honestly, where the graph does *not* help: a basin-aware refinement of the graph performs within noise of a plain nearest-neighbour graph, and a station-level connectivity effect is directionally robust but statistically underpowered.

Contributions.

- We build a reproducible, leakage-safe benchmark for buoy wave-field reconstruction and forecasting from a national NDBC coverage audit, releasing a hide-and-recover protocol that separates a spatial regime (whole-station outages) from a temporal regime (random and block missingness).
- We show that a graph imputation model (GRIN) wins wherever missingness is structured (whole-station outages and contiguous blocks), and we isolate the graph’s specific contribution with a per-station comparison against a strong non-spatial attention model.
- We benchmark eight forecasters, from persistence and autoregression to transformer SOTA (PatchTST, iTransformer) and three spatiotemporal graph models, and show that the graph family forms a distinct upper tier at horizons of 6 h or more, while the temporal models, however sophisticated, cluster together far below.
- We give a *physical explanation* for that advantage: the buoy network recovers the deep-water dispersion relation (cross-correlation lags track the group-velocity prediction d/c_g , $\rho = 0.80$), we map the basin-scale swell corridors it implies, and we show the graph’s edge is its ability to use an upstream buoy as a causal preview of arriving swell.
- We carry the forecasts through to wave power and show that the H^2 nonlinearity *amplifies* the graph model’s advantage, with method rankings invariant to the period-conversion coefficient.
- We report two negative or underpowered results (a within-noise basin-aware graph and a suggestive-only connectivity effect), so the benchmark is not oversold.

2 Related Work

2.1 Time-series imputation

The simplest fixes, mean fill, forward fill, and linear interpolation, treat each series independently and ignore both cross-station structure and uncertainty. Classical spatial interpolation (inverse-

distance weighting, kriging) uses neighbours but not temporal dynamics [CITATION NEEDED: spatial interpolation / kriging]. Recurrent models learn temporal dependencies directly: BRITS imputes multivariate series with a bidirectional recurrent network and treats missing values as trainable quantities [4]. Attention-based models replace recurrence with self-attention: SAITS uses a joint reconstruction and imputation objective and is a strong, purely temporal baseline [8]. Diffusion models such as CSDI cast imputation as conditional generation and provide calibrated uncertainty [16]. A recent survey catalogues this progression [19]. What these share is a weak or absent spatial prior. Graph imputation closes that gap: GRIN combines a message-passing graph with recurrent units so that a missing value at one node is filled from its neighbours as well as its own history [6]. This is the model we adopt for imputation.

2.2 Spatiotemporal graph learning

Representing sensors as graph nodes and learning over space and time has become standard for networked signals. Diffusion-convolutional recurrent networks model traffic as diffusion on a graph [11], and Graph WaveNet couples a learned adjacency with dilated temporal convolutions to capture long-range dependencies efficiently [20]. These methods motivate a graph approach for buoys, where the “sensors” are stations and the edges encode ocean proximity. We use Graph WaveNet as our forecasting model.

2.3 Wave forecasting

Operational wave prediction is dominated by physics-based spectral models that solve the wave-action balance over a grid [2, 17]. These are accurate but computationally heavy and require boundary and wind forcing. Data-driven forecasting offers a lightweight complement, and for short horizons the hardest baseline to beat is persistence, tomorrow looks like today. We therefore benchmark against persistence and a linear autoregressive model and ask whether a graph model adds skill on top.

2.4 Wave-energy estimation

The deep-water wave energy flux (power per unit crest length) scales with the square of significant wave height and linearly with the energy period [3]. Because operational archives report a mean or zero-crossing period rather than the energy period, a documented conversion is needed; Cahill and Lewis [3] analyse these period ratios and their effect on computed wave power. Resource assessments build on such conversions: Guillou [10], using NOAA/NDBC deep-water buoys, finds that an energy-period formulation accurately approximates deep-water wave power, whereas using the peak period with a default coefficient overestimates it, directly motivating our choice to derive T_e from the reported average period rather than a peak period. The applied value of better height and period forecasts is therefore best judged in energy units, which is where we end our study.

3 Data and Study Area

We use standard meteorological records from the U.S. National Data Buoy Center [13]. We first ran a national coverage audit over all active stations, scanning combinations of minimum temporal coverage and minimum record length and stripping NDBC missing-value sentinels before counting. From 872 audited stations we selected 140 that meet a coverage-and-longevity threshold over 2021–2025, yielding an hourly tensor of 43,824 time steps across the station panel used throughout

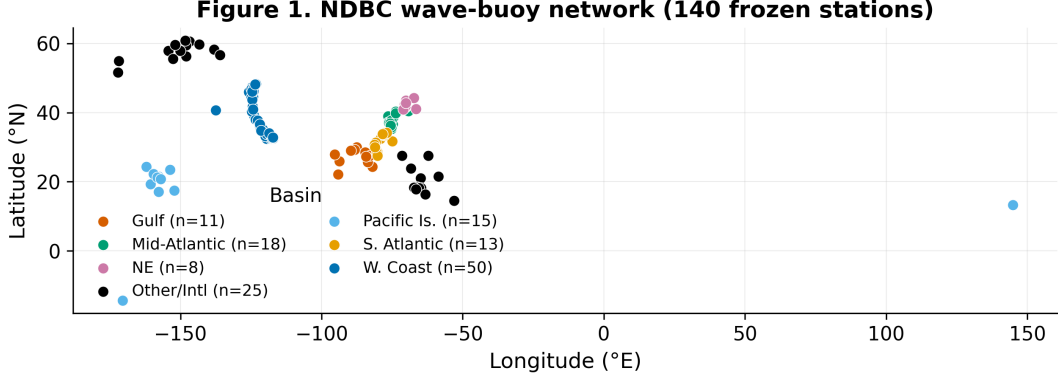


Figure 1: Selected NDBC stations across U.S. ocean basins after the national coverage audit and metadata corrections. Nodes are buoys; the panel defines the spatial domain for the graph.

(Figure 1). The connectivity analysis in Section 6 is limited to the station-outage victim stations, of which there are 38 with a scorable WVHT recovery and 33 with a scorable APD recovery (five victims do not report APD and so cannot be scored on it). These per-target sample sizes are much smaller than the full panel.

We correct two classes of metadata error discovered during the audit. First, a basin misassignment near the Florida–Atlantic boundary was fixed by adjusting the basin box so that Atlantic stations are not labelled Gulf. Second, a station whose archived coordinate placed it near Japan while its name placed it southwest of Honolulu was corrected to its documented Pacific location. Both corrections are recorded in configuration and applied before the graph is built.

Our targets are significant wave height (WVHT, m) and average wave period (APD, s); dominant wave period (DPD) is retained as a predictor feature. NDBC reports wave fields a few minutes past the hour rather than exactly on the hour, so we harmonize timestamps by taking the first valid reading within each hour, producing a clean hourly cadence. Figures 2–6 summarize missingness, coverage, target distributions, diurnal and seasonal cycles, and the height–period relationship that the models exploit.

4 Methodology

4.1 Spatiotemporal tensor

We arrange the data as a tensor $X \in \mathbb{R}^{T \times N \times F}$ of T hourly steps, N stations, and F features (the targets plus retained covariates). A binary mask M of the same shape marks genuinely observed entries. All models see X and M ; no model is allowed to treat an imputed or padded value as if it were observed.

4.2 Graph construction

We connect stations with a k -nearest-neighbour graph in geographic distance ($k = 8$). Edge weights follow a self-tuning Gaussian kernel, $w_{ij} = \exp(-d_{ij}^2/(\sigma_i\sigma_j))$, where σ_i is the distance from station i to its k -th neighbour. This local scaling avoids the numerical underflow that a single global bandwidth produces when station spacing varies by orders of magnitude across basins. We also build a *basin-aware* variant that zeros edges crossing basin boundaries, to test whether forbidding

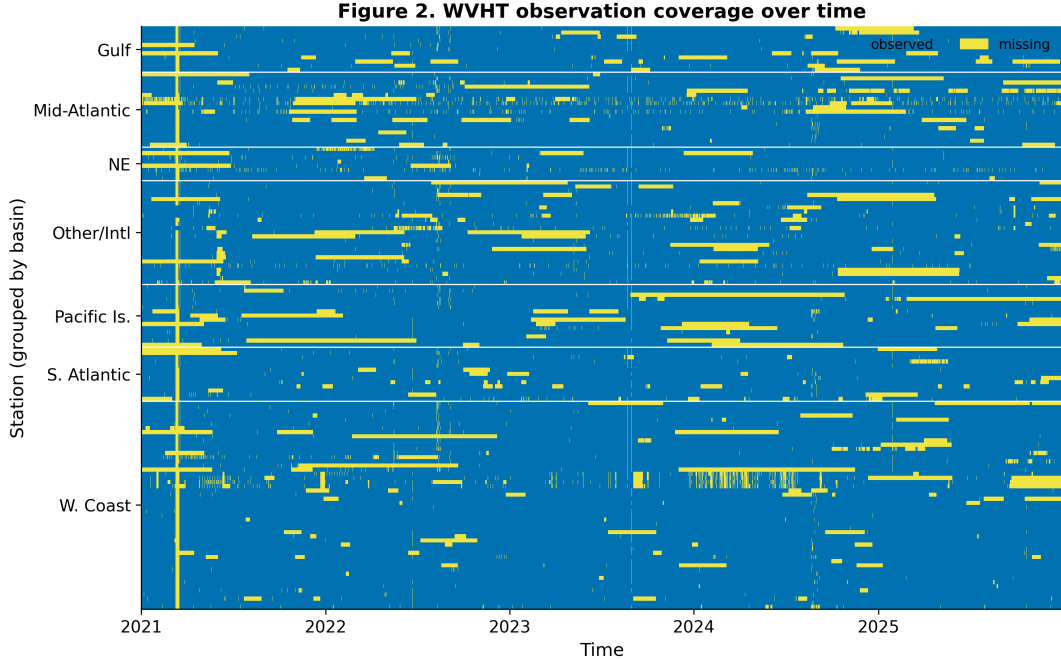


Figure 2: Missingness structure across stations and time. Gaps are both random (scattered hourly dropouts) and structured (whole-station outages spanning long intervals), motivating separate evaluation regimes.

cross-basin message passing helps. Figures 7 and 8 show the empirical correlation–distance decay and the resulting graph.

4.3 Leakage-safe hide-and-recover protocol

For imputation we hide observed entries, ask each model to recover them, and score only on the hidden-but-known cells. We define two regimes. The *spatial* regime removes whole stations for extended intervals (station-outage), so a model can only recover a victim from its neighbours and its own surrounding history. The *temporal* regime removes values within otherwise-present stations, either completely at random (MCAR) or in contiguous blocks, at 10%, 30%, and 50% rates. Scoring is confined to truly held-out cells, and no target leaks through the covariates.

4.4 Imputation models

We compare (i) simple baselines, mean fill, forward fill, and linear interpolation; (ii) spatial baselines, inverse-distance weighting (IDW) and a combined spatial-plus-temporal interpolator; (iii) SAITS, a self-attention imputer [8]; (iv) BRITS, a bidirectional recurrent imputer [4]; and (v) GRIN, a graph recurrent imputer [6]. GRIN is bidirectional, which is legitimate for imputation because the task is reconstruction of an interior gap from both sides.

4.5 Forecasting models and the causal-input design

Forecasting must respect the arrow of time. We use an input window of $L = 24$ hours and predict targets h steps ahead. Baselines are persistence (repeat the last value), seasonal-naive (repeat the value one day earlier), and a per-station autoregressive model, AR(24), fit by least squares as a direct h -step predictor. Our graph model is Graph WaveNet [20].

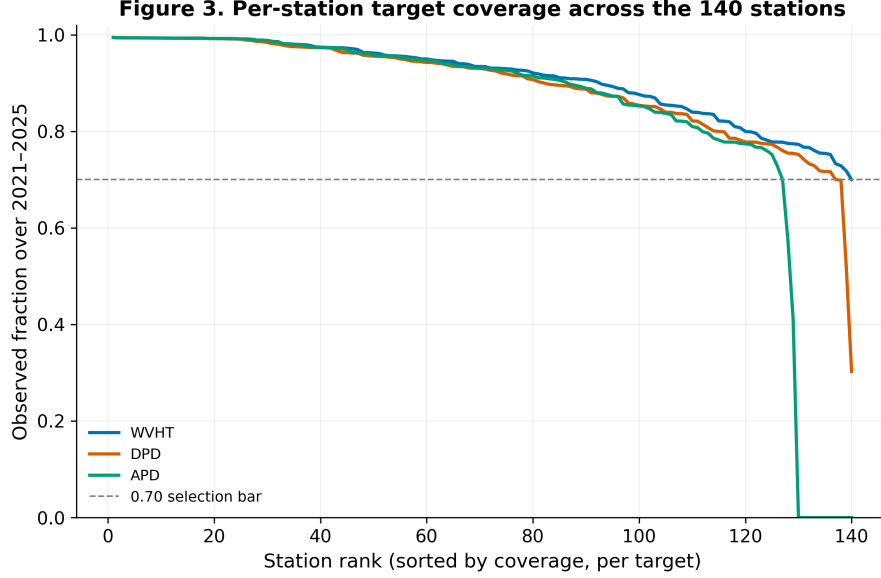


Figure 3: Distribution of per-station temporal coverage over the study window, used to set the selection threshold.

To avoid leakage we separate training and test information sources. The model is *trained* on a graph-completed tensor, missing training entries filled by GRIN, where bidirectional filling is acceptable because it is confined to the training period. At *test* time the model reads only causal, forward-filled inputs that use past information alone, and we evaluate only at origins where the entire input window and the target at $t + h$ are genuinely observed (a valid-origin mask). This keeps future information out of every test forecast.

4.6 Wave-energy computation

We convert forecasts to deep-water wave power [10],

$$P = \frac{\rho g^2}{64\pi} H_{m0}^2 T_e \approx 0.49 H^2 T_e \quad (\text{kW/m}), \quad (1)$$

with H in metres and T_e in seconds. Because NDBC reports APD rather than the energy period T_e , we set $T_e = \alpha \cdot \text{APD}$ with $\alpha = 1.12$, a wave-period ratio consistent with the empirical T_e/T_{02} values reported by Cahill and Lewis [3];¹ α is an explicit configuration parameter. True energy uses observed WVHT and APD at $t + h$, and a station-hour is scored only where *both* targets are genuinely observed. Since α multiplies every method’s energy identically, method rankings and persistence-normalized skill are invariant to α ; only the absolute kW/m magnitudes depend on it.

5 Experimental Setup

Splits. We train on 2021–2023, validate on 2024, and test on 2025, so no test period is ever seen during fitting.

¹This treats NDBC’s average wave period (APD) as equivalent to the average zero-crossing period T_{02} used in that ratio, an assumption to confirm with co-authors, not an established identity.

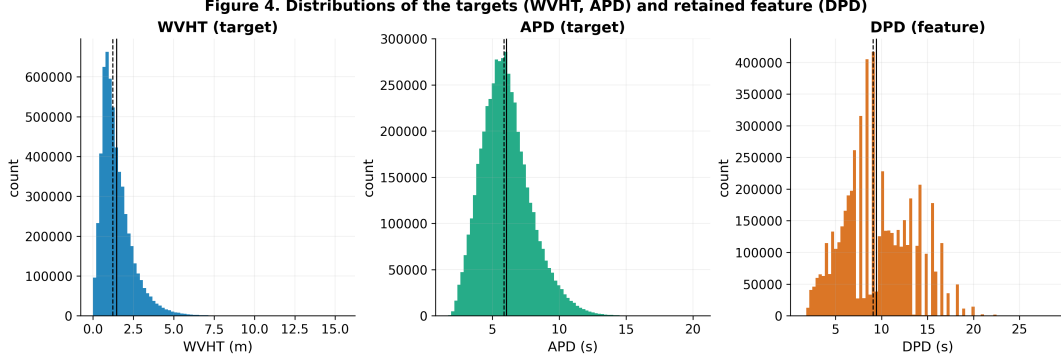


Figure 4: Pooled distributions (all stations $\times$ hours) of the two modelled targets, significant wave height (WVHT) and average wave period (APD), and the retained predictor feature, dominant wave period (DPD). DPD is a model input, not a target. Solid and dashed lines mark the mean and median.

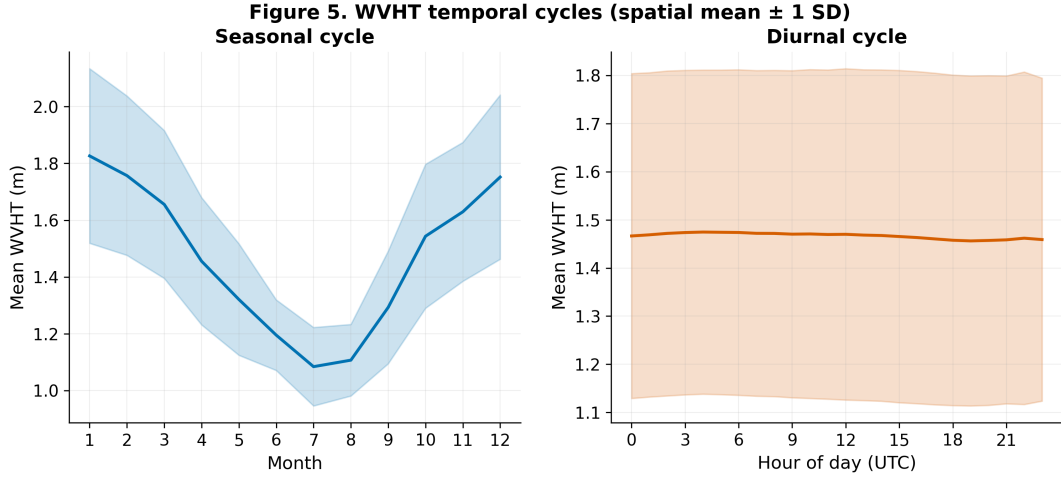


Figure 5: Diurnal and seasonal cycles in the targets, showing the temporal structure available to single-station baselines.

Masking. MCAR and block masks are applied at 10/30/50% rates; station-outage masks remove a fixed set of victim stations. To measure variability we repeat each masked evaluation over three model seeds and, for the deep models, average within a run before taking the standard deviation across seeds, so error bars reflect model-training variability rather than mask resampling.

Metrics. We report MAE and RMSE in physical units (m for WVHT, s for APD, kW/m for power) and a persistence-normalized skill, $\text{skill} = 1 - \text{MAE}_{\text{model}} / \text{MAE}_{\text{persistence}}$, so that positive values mean “better than persistence.”

Forecasting models. Besides persistence and the per-station AR(24), we evaluate three temporal deep models, DLinear [21], PatchTST [14] and iTransformer [12], and three spatiotemporal graph models, Graph WaveNet [20], DCRNN [11] and AGCRN [1]. All are direct multi-horizon forecasters: they read a 24-hour input window across all stations and predict 24 hours ahead in one shot, so every shorter horizon is read from the same forecast. All share the leakage-safe protocol (trained on the GRIN-completed 2021–2024 tensor, scored on causal forward-filled test inputs at the same origins as the baselines).

Hyperparameters and compute. Deep models are seeded (seed 0) and checkpointed, with best weights saved on every validation improvement. GRIN is trained with a capped budget (up to 50

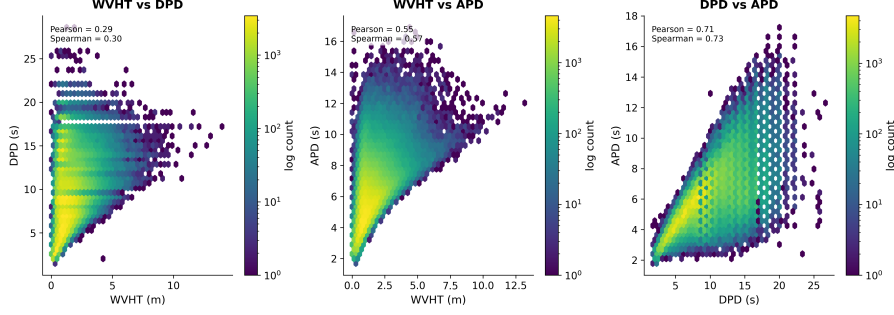


Figure 6: Joint structure of height and period, the relationship the wave-power computation depends on.

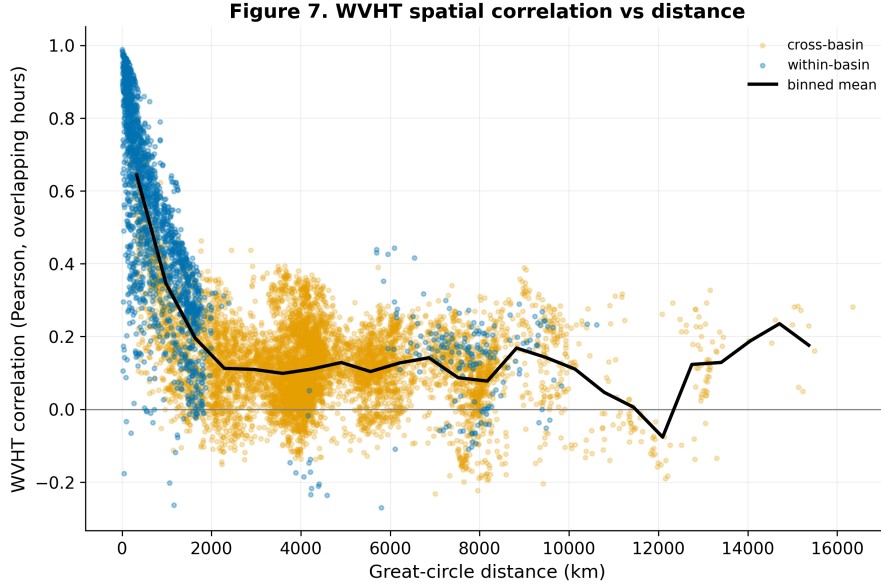


Figure 7: Inter-station target correlation as a function of separation distance. Correlation decays with distance, justifying a distance-weighted neighbourhood.

epochs, early stopping with patience 10). All forecasters are trained on CPU: Graph WaveNet’s channels-last convolution is unsupported on the Apple MPS backend, and we observed that sustained MPS load both degrades throughput and, for the full-tensor imputation pass, exhausts GPU memory, so CPU is used for reproducibility. For compute tractability the transformer and graph forecasters are trained on a 1-in-3 subsample of forecast origins (capped at 20 epochs); DLinear and the classical baselines use every origin. Adjacent hourly origins are highly correlated, so this subsampling has little effect on the fitted models. All runs use a fixed software stack (`ts1` [5], `PyPOTS` [7], `PyTorch` [15], `PyTorch Geometric` [9]).

6 Results

6.1 Imputation

Table 1 reports imputation MAE for both targets in three representative regimes. GRIN achieves the lowest error wherever the missing gaps are *structured*. Under whole-station outages it recovers WVHT at 0.346 ± 0.007 m, far below the best spatial baseline (IDW, 0.443 m) and the spatial-

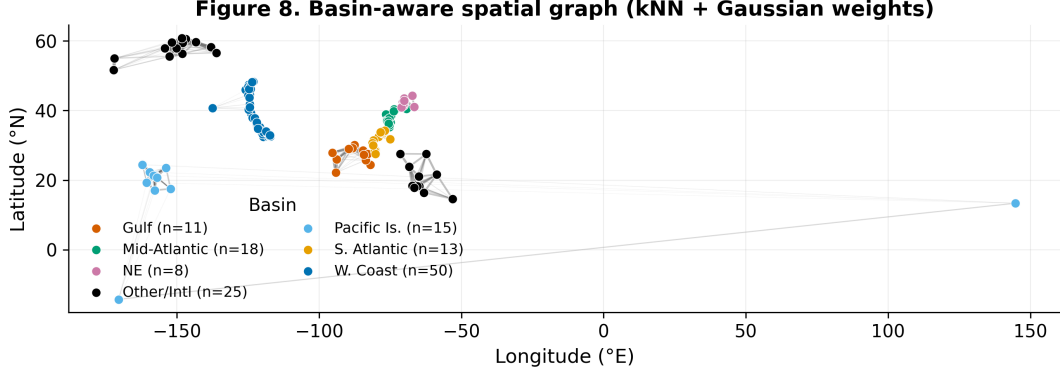


Figure 8: The k -nearest-neighbour Gaussian station graph. Edge shading encodes the self-tuning kernel weight; the basin-aware variant removes cross-basin edges.

Table 1: Imputation mean absolute error (lower is better). Station-outage is the spatial regime; MCAR-50% and Block-50% are temporal regimes. GRIN values are mean $\pm$ standard deviation over three model seeds. Best per column in bold.

Method	Station-outage		MCAR-50%		Block-50%	
	WVHT	APD	WVHT	APD	WVHT	APD
Mean fill	0.458	0.955	0.476	0.997	0.445	0.968
Forward fill	0.637	1.303	0.111	0.306	0.702	1.410
Linear interp	0.637	1.303	0.074	0.211	0.557	1.186
Spatial IDW	0.443	0.765	0.430	0.844	0.437	0.889
Spatial+temporal	0.426	0.846	0.229	0.462	0.392	0.857
SAITS	0.730	1.826	0.075	0.209	0.595	0.827
BRITS	0.725	1.565	0.124	0.351	0.600	0.898
GRIN	0.346$\pm$0.007	0.726$\pm$0.033	0.082 $\pm$ 0.002	0.228 $\pm$ 0.002	0.373$\pm$0.021	0.704$\pm$0.039

plus-temporal interpolator (0.426 m), and well below the purely temporal deep models (SAITS 0.730 m, BRITS 0.725 m) whose single-station design cannot see a station that is entirely absent. It is also best under 50% contiguous-block missingness for both targets. The picture changes only under 50% *completely random* missingness: isolated single-hour gaps are easy to interpolate, so the temporal methods lead. Linear interpolation (WVHT 0.074 m) is best by a hair over the attention model SAITS (0.075 m), and GRIN (0.082 m) is competitive but behind, the graph adds little when time alone suffices. The message is not that GRIN dominates everywhere, but that it is best exactly where spatial or temporal structure must be exploited, and merely competitive where a gap is trivial to fill.

Robustness across missingness rates. Tables 2 and 3 extend the comparison to five missingness rates (5–50%) for both the random (MCAR) and contiguous-block regimes, so the ranking is not read off a single rate. The pattern is stable across the whole range. Under MCAR, where a hidden hour usually sits between two observed neighbours in time, the temporal methods win at every rate: the attention model SAITS is best at almost every rate (linear interpolation edges it only at the highest WVHT rate). GRIN trails the temporal leaders here, and the spatial-only interpolators trail furthest. Time, not space, is the easy axis. Under contiguous blocks the ranking reverses cleanly: GRIN is best at every block rate for *both* targets, because a longer temporal gap is exactly where a station must borrow from its neighbours. The single-paradigm floor and the

Table 2: Imputation MAE for **WVHT** (m) across missingness rates, MCAR and Block at 5/10/20/30/50%. GRIN is the basin-aware graph (mean over three model seeds). Lower is better; best per column in bold.

Method	MCAR					Block				
	5%	10%	20%	30%	50%	5%	10%	20%	30%	50%
Mean fill	0.476	0.476	0.476	0.476	0.476	0.470	0.474	0.465	0.451	0.445
Forward fill	0.087	0.088	0.092	0.097	0.111	0.557	0.599	0.586	0.628	0.702
Linear interp	0.065	0.065	0.067	0.068	0.074	0.460	0.520	0.505	0.504	0.557
Spatial IDW	0.371	0.376	0.387	0.399	0.430	0.368	0.366	0.373	0.381	0.437
Spatial+temporal	0.200	0.202	0.207	0.213	0.229	0.337	0.351	0.357	0.354	0.392
SAITS	0.052	0.053	0.057	0.062	0.075	0.628	0.643	0.639	0.618	0.595
BRITS	0.079	0.082	0.089	0.098	0.124	0.688	0.687	0.667	0.635	0.600
GRIN	0.065	0.066	0.068	0.071	0.082	0.325	0.333	0.339	0.341	0.373

Table 3: Imputation MAE for **APD** (s) across missingness rates, MCAR and Block at 5/10/20/30/50%. GRIN is the basin-aware graph (mean over three model seeds). Lower is better; best per column in bold.

Method	MCAR					Block				
	5%	10%	20%	30%	50%	5%	10%	20%	30%	50%
Mean fill	0.997	0.998	0.999	0.997	0.997	1.002	0.989	0.981	0.969	0.968
Forward fill	0.240	0.244	0.255	0.268	0.306	1.202	1.213	1.258	1.281	1.410
Linear interp	0.177	0.179	0.184	0.191	0.211	1.059	1.084	1.094	1.087	1.186
Spatial IDW	0.752	0.760	0.777	0.795	0.844	0.768	0.772	0.783	0.789	0.889
Spatial+temporal	0.408	0.412	0.422	0.433	0.462	0.761	0.770	0.778	0.785	0.857
SAITS	0.148	0.152	0.162	0.174	0.209	0.819	0.823	0.835	0.827	0.827
BRITS	0.215	0.225	0.249	0.276	0.351	0.947	0.934	0.928	0.913	0.898
GRIN	0.178	0.181	0.189	0.198	0.228	0.616	0.615	0.627	0.640	0.704

graph’s advantage are therefore properties of the regime, not artefacts of one masking rate.

Isolating the graph’s contribution. Comparing GRIN against SAITS station by station isolates the effect of adding spatial message passing to a strong temporal model. Averaged over victim stations, GRIN reduces station-outage error by about 0.38 m in WVHT and 0.86 s in APD relative to SAITS. This gap is the graph’s specific contribution: the two models share the deep-learning machinery, and only GRIN uses neighbours.

Does station connectivity predict recovery? If the graph is the reason GRIN recovers outages, better-connected stations should be easier to recover. Table 4 tests this with Spearman correlations between three connectivity measures and recovery error, in raw form, after normalizing by each station’s own variability, and as the GRIN-minus- SAITS advantage. The signs are consistent with the hypothesis, higher degree and denser edge weight tend to lower normalized error, and several coefficients reach significance ($*p < 0.05$). We deliberately do not overclaim: the sample is small ($n = 38$ victim stations for WVHT and $n = 33$ for APD), so we read this as *suggestive*, directional evidence rather than a firm law. Figure 9 visualizes the relationship.

A within-noise negative result: basin-aware edges. We expected that forbidding cross-basin message passing would help, since distant basins share little wave physics. It does not. Table 5 compares the basin-aware graph against the plain k -nearest-neighbour graph; in 24 of 24

Table 4: Spearman correlation between station connectivity and station-outage recovery. Negative values mean better-connected stations are recovered more accurately. “Normalized” divides error by the station’s own variability; “advantage” is GRIN minus SAITS. $*p < 0.05$. Samples are small ($n = 38$ stations for WVHT, $n = 33$ for APD), so these are suggestive.

Target	Connectivity measure	Raw	Normalized	Advantage
WVHT	degree	-0.36*	-0.53*	+0.21
WVHT	neighbour dist.	+0.59*	+0.29	-0.19
WVHT	edge-weight sum	-0.46*	-0.52*	+0.16
APD	degree	+0.26	-0.35*	+0.16
APD	neighbour dist.	-0.46*	-0.01	-0.47*
APD	edge-weight sum	+0.40*	-0.20	+0.38*

Table 5: Basin-aware versus plain k -NN graph (imputation MAE, mean $\pm$ standard deviation over three model seeds). “Exceeds noise” is yes only when the advantage is larger than the combined seed standard deviation.

Regime	Target	Basin-aware	Plain k -NN	Advantage	Seed SD	Exceeds noise
Station-outage	WVHT	0.346 $\pm$ 0.007	0.340 $\pm$ 0.013	-0.006	0.015	no
Station-outage	APD	0.726 $\pm$ 0.033	0.745 $\pm$ 0.052	+0.018	0.062	no
MCAR-50%	WVHT	0.082 $\pm$ 0.002	0.083 $\pm$ 0.002	+0.001	0.003	no
MCAR-50%	APD	0.228 $\pm$ 0.002	0.228 $\pm$ 0.002	+0.000	0.003	no
Block-50%	WVHT	0.373 $\pm$ 0.021	0.363 $\pm$ 0.015	-0.010	0.026	no
Block-50%	APD	0.704 $\pm$ 0.039	0.716 $\pm$ 0.032	+0.012	0.050	no

target-regime cells the difference is smaller than the combined model-seed noise. We report this as a useful simplification: the plain graph is adequate, and the extra basin bookkeeping is unnecessary.

6.2 Swell propagation and the graph mechanism

Why do the graph models win? We show the reason is physical: the buoy network encodes swell propagation, and the graph’s advantage is the ability to exploit it. Ocean swell travels between buoys at the deep-water group velocity $c_g = gT/4\pi$, so a wave event at an upstream buoy reaches a downstream buoy after a lag $\tau = d/c_g$. For every within-basin buoy pair we take the lag that maximises the cross-correlation of their significant-wave-height series and compare it to this prediction (Figure 10). The observed lags track the physics closely: Spearman $\rho = 0.80$ over 1,427 within-basin pairs, and 99% of pairs respect the causal bound $|\tau^*| \leq d/c_g$, so the correlation almost never propagates faster than the group velocity. The agreement is coherent within every basin and strongest along the West Coast ($\rho = 0.89$), where long Pacific swell dominates. The network has recovered the deep-water dispersion relation from data alone.

The lead-lag structure defines a directed *swell-propagation graph* (Figure 11): 758 upstream-to-downstream edges over 133 of the 140 buoys, with a median propagation lag of 5 hours. These are data-derived maps of basin-scale swell corridors.

This propagation is the mechanism behind the two-tier forecasting result below. To forecast a buoy h hours ahead, an upstream buoy roughly h hours away is a *causal preview*: its already-observed state is the swell that will arrive at the target. It explains why forecast skill *grows with horizon* (Figure 12), since longer horizons make more distant upstream buoys usable as previews, and, in a controlled test, propagation-lag-aligned upstream features improve 24 h WVHT forecasts over same-time neighbour features. The spatial graph wins because it reads the wave field the way

Table 6: Forecast MAE and persistence-normalized skill (in parentheses) at horizons of 1 to 24 hours. Persistence is the reference, so its skill is zero by definition. Higher skill is better.

Method	WVHT (m)					APD (s)				
	$h=1$	$h=3$	$h=6$	$h=12$	$h=24$	$h=1$	$h=3$	$h=6$	$h=12$	$h=24$
Persistence	0.086	0.138	0.205	0.296	0.416	0.236	0.386	0.550	0.724	0.884
AR(24)	0.083 (+0.038)	0.133 (+0.039)	0.197 (+0.042)	0.280 (+0.054)	0.378 (+0.090)	0.229 (+0.030)	0.369 (+0.045)	0.511 (+0.072)	0.654 (+0.096)	0.789 (+0.107)
DLinear	0.085 (+0.017)	0.135 (+0.020)	0.200 (+0.024)	0.285 (+0.039)	0.386 (+0.071)	0.233 (+0.012)	0.380 (+0.015)	0.535 (+0.028)	0.692 (+0.043)	0.828 (+0.063)
PatchTST	0.083 (+0.030)	0.133 (+0.038)	0.195 (+0.048)	0.277 (+0.064)	0.379 (+0.089)	0.231 (+0.018)	0.375 (+0.029)	0.525 (+0.046)	0.676 (+0.066)	0.815 (+0.078)
iTransformer	0.085 (+0.010)	0.133 (+0.038)	0.194 (+0.053)	0.274 (+0.074)	0.375 (+0.097)	0.232 (+0.014)	0.372 (+0.037)	0.517 (+0.061)	0.663 (+0.084)	0.805 (+0.089)
GraphWaveNet	0.083 (+0.039)	0.124 (+0.103)	0.177 (+0.138)	0.249 (+0.158)	0.347 (+0.166)	0.223 (+0.055)	0.349 (+0.096)	0.478 (+0.132)	0.607 (+0.161)	0.745 (+0.157)
DCRNN	0.084 (+0.019)	0.126 (+0.091)	0.179 (+0.128)	0.252 (+0.150)	0.350 (+0.159)	0.230 (+0.024)	0.352 (+0.088)	0.478 (+0.130)	0.605 (+0.163)	0.748 (+0.153)
AGCRN	0.089 (-0.040)	0.129 (+0.067)	0.181 (+0.119)	0.248 (+0.163)	0.333 (+0.199)	0.240 (-0.016)	0.359 (+0.070)	0.482 (+0.124)	0.595 (+0.177)	0.716 (+0.189)

the ocean actually propagates it.

6.3 Forecasting

Table 6 reports forecast MAE and persistence-normalized skill across five horizons, from 1 to 24 hours, for eight methods: persistence and a per-station autoregression (AR(24)); three strong temporal deep models (DLinear, PatchTST, iTransformer); and three spatiotemporal graph models (DCRNN, AGCRN, and Graph WaveNet). Three findings stand out.

First, persistence is extremely strong at very short range: at 1 hour its WVHT error is only 0.086 m, and the best learned model improves on it by only +0.039, so we report honestly that a learned model adds little at 1 hour. All skill grows with horizon as the sea state has time to evolve.

Second, and most important, the methods separate into two clean tiers by *paradigm*. At every horizon of 6 hours or more, all three graph models beat all three temporal deep models and the autoregression. At 24 hours no temporal method exceeds +0.097 WVHT skill, while no graph model falls below +0.159: the tiers do not overlap. The strongest temporal SOTA models, PatchTST and iTransformer, are no better than the classical AR(24). The decisive ingredient is spatial information, not temporal architecture; a model that can read a buoy’s neighbours forecasts it far better than one that cannot, however sophisticated its treatment of time.

Third, the graph advantage is a property of the graph *family*, not of one architecture. Graph WaveNet (fixed k -NN graph), DCRNN (diffusion on that graph), and AGCRN (a *learned* adjacency) all reach the upper tier. AGCRN is best at long range (+0.199 WVHT skill at 24 hours) but worst at 1 hour (-0.040, below persistence): a learned graph needs a longer context to pay off. Figure 12 shows the two tiers and the widening gap as the horizon grows.

6.4 Wave energy

Table 7 carries the forecasts through Equation 1 to wave power, scored on the same leakage-safe origins where both WVHT and APD are observed, at every horizon from 1 to 24 hours. The two-tier split from the forecasts not only persists but *widens* in energy space. At 24 hours every graph model exceeds +0.226 energy skill while no temporal method reaches +0.173, and AGCRN leads at +0.266. Graph WaveNet’s own energy skill rises with horizon to +0.217 at $h=12$ and +0.235 at $h=24$, larger than its height and period skill. The reason is the H^2 nonlinearity in Equation 1: a model that forecasts wave height more accurately earns a squared benefit in energy, so the graph model’s edge on height is magnified in power. Because α scales all methods equally, these rankings and skills do not depend on the period-conversion coefficient. Figure 13 summarizes the result.

Table 7: Wave-energy forecast MAE (kW/m) and persistence-normalized skill (in parentheses) at horizons of 1 to 24 hours, scored where both WVHT and APD are observed. Rankings and skills are invariant to the period coefficient α .

Method	$h=1$	$h=3$	$h=6$	$h=12$	$h=24$
Persistence	1.752	2.579	3.764	5.468	7.691
AR(24)	1.658 (+0.054)	2.441 (+0.054)	3.489 (+0.073)	4.863 (+0.111)	6.434 (+0.164)
DLinear	1.697 (+0.031)	2.486 (+0.036)	3.569 (+0.052)	4.972 (+0.091)	6.581 (+0.144)
PatchTST	1.671 (+0.046)	2.433 (+0.057)	3.450 (+0.083)	4.792 (+0.124)	6.402 (+0.168)
iTransformer	1.748 (+0.002)	2.473 (+0.041)	3.475 (+0.077)	4.778 (+0.126)	6.358 (+0.173)
GraphWaveNet	1.669 (+0.048)	2.278 (+0.117)	3.113 (+0.173)	4.284 (+0.217)	5.882 (+0.235)
DCRNN	1.665 (+0.050)	2.289 (+0.112)	3.148 (+0.164)	4.348 (+0.205)	5.949 (+0.226)
AGCRN	1.811 (-0.034)	2.383 (+0.076)	3.192 (+0.152)	4.274 (+0.219)	5.648 (+0.266)

7 Discussion

The results tell one coherent story: a spatiotemporal graph helps most where spatial structure matters most. In imputation this is literal. When a whole station is missing, its own history is gone and only neighbours can help; the graph model is the only method that is best here, and the per-station GRIN-minus- SAITS gap shows the improvement comes specifically from message passing, not from deep learning in general. When gaps are isolated and random, the advantage disappears (an isolated missing hour is trivial to interpolate), exactly as one would expect if the graph’s value is the recovery of *structured* loss.

In forecasting the same logic explains both the null and the positive result. At 1 hour the sea state barely changes, so persistence is near-optimal and there is little spatial information to add. As the horizon lengthens, swell propagation between buoys becomes informative, and the graph model converts that propagation into skill. Carrying forecasts into energy sharpens the picture further: because power squares wave height, the graph model’s height advantage is amplified, and its energy skill exceeds its height skill. For a practitioner siting a wave-energy converter, this is the number that matters, and it is where the graph wins by the widest margin.

We are careful about what we do not show. The basin-aware graph performs within model-seed noise of the plain graph; rather than bury this, we present it as a simplification, basin bookkeeping is unnecessary here. The connectivity effect has the expected sign and some significant coefficients, but with only $n = 38$ victim stations for WVHT (and $n = 33$ for APD) it is underpowered, and we label it suggestive.

Limitations. Our evaluation is confined to two targets (WVHT and APD) over one national archive and one multi-year window; other basins, variables, and periods may behave differently. The energy computation uses a deep-water approximation and a single period-conversion coefficient; although rankings are invariant to that coefficient, absolute power magnitudes are not, and shallow-water sites would need a different flux formula that accounts for water depth [18]. Deep models were trained under capped budgets on CPU for reproducibility, not tuned for maximum accuracy, so the reported skills are conservative. Finally, we benchmark against statistical baselines, not operational physics-based wave models [2, 17]; a hybrid comparison is left to future work.

8 Conclusion

We asked whether one modelling idea, a spatiotemporal graph that shares information between buoys, improves the full applied pipeline for ocean waves. Across imputation, forecasting, and

wave-energy estimation, the answer is yes, and consistently for the same reason: the graph pays off where spatial structure is available to exploit. It is the best imputer wherever missingness is structured (whole-station outages and contiguous blocks), while isolated random gaps are easy enough that simple interpolation suffices; it adds real 12–24 h forecast skill over persistence and autoregression; and it wins by its widest margin on wave power, where the H^2 nonlinearity amplifies its superior height forecasts. We also report where it does not help (a within-noise basin-aware graph and an underpowered connectivity effect), so that the benchmark is honest about its limits. Future work should extend the panel, compare against operational wave models, and add calibrated uncertainty to the forecasts.

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Figure 9. Confound-controlled: connectivity vs GRIN outage benefit (full-test station-outage)

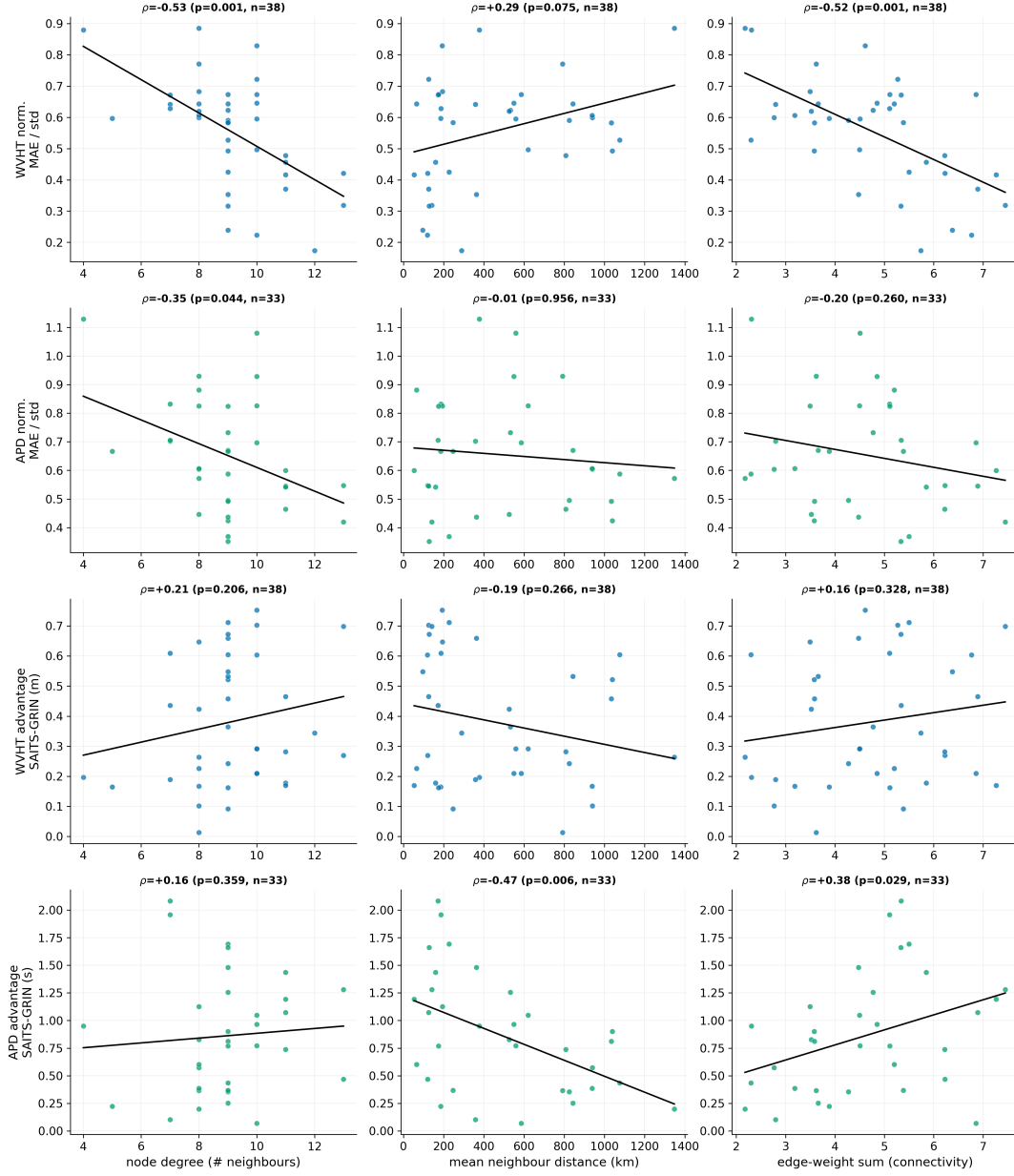


Figure 9: Station-outage recovery error against graph connectivity. Better- connected stations tend to be recovered more accurately, but the trend is based on a small sample and is reported as suggestive.

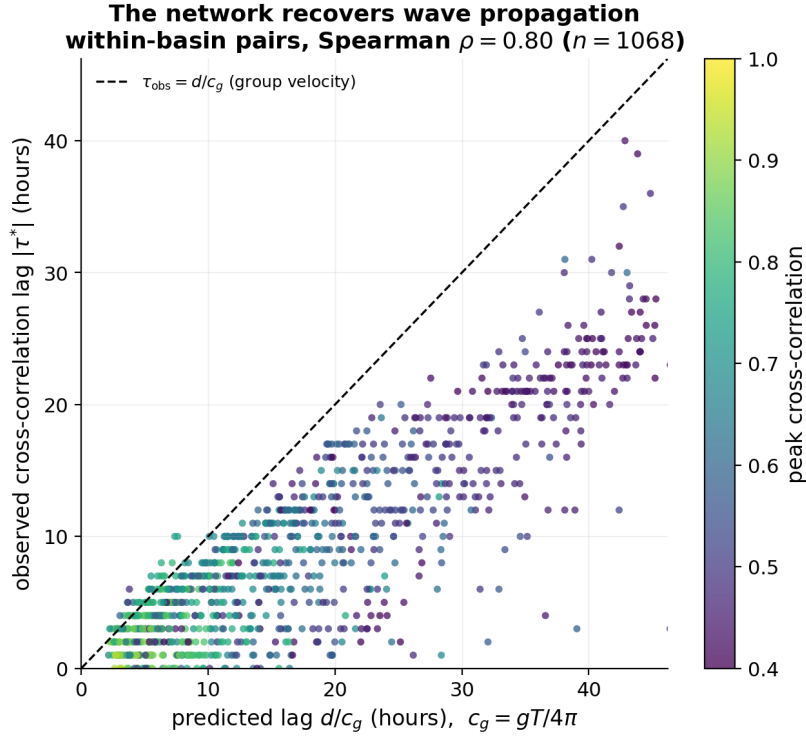


Figure 10: The buoy network recovers wave propagation. For within-basin pairs, the observed cross-correlation lag $|\tau^*|$ tracks the deep-water group-velocity prediction d/c_g ; points colour by coupling strength. Nothing lies above the diagonal: the correlation does not outrun the group velocity.

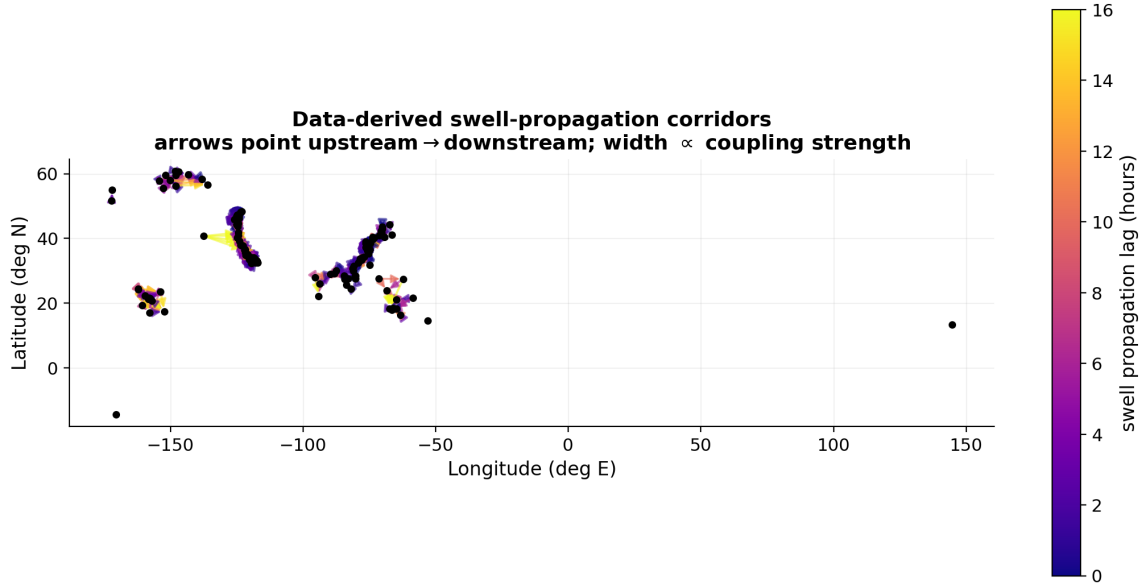


Figure 11: Data-derived swell-propagation corridors. Each arrow runs from the upstream (leading) buoy to the downstream buoy of a strongly-coupled within-basin pair, coloured by propagation lag and drawn with width proportional to coupling strength.

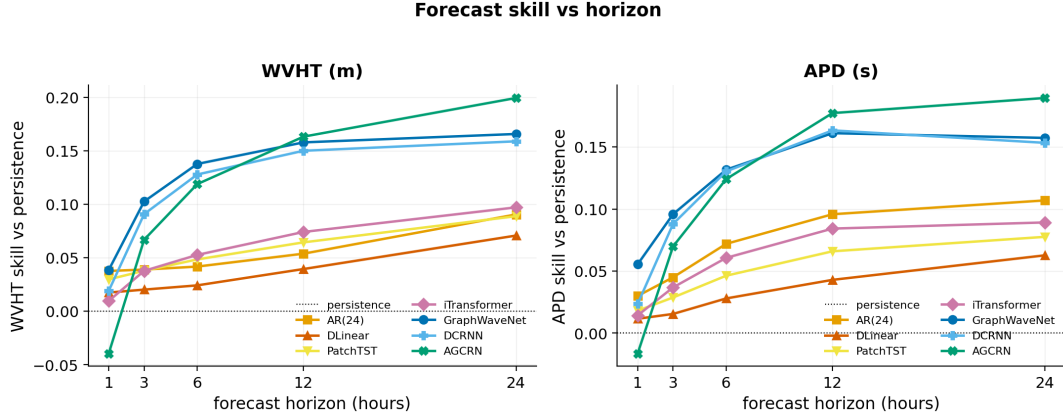


Figure 12: Forecast skill versus horizon for the eight methods. The three spatiotemporal graph models (Graph WaveNet, DCRNN, AGCRN) form an upper tier that separates from the four temporal methods (AR(24), DLinear, PatchTST, iTransformer) at every horizon of 6 hours or more; the gap widens with horizon. AGCRN's learned graph overtakes at 24 hours but is weakest at 1 hour.

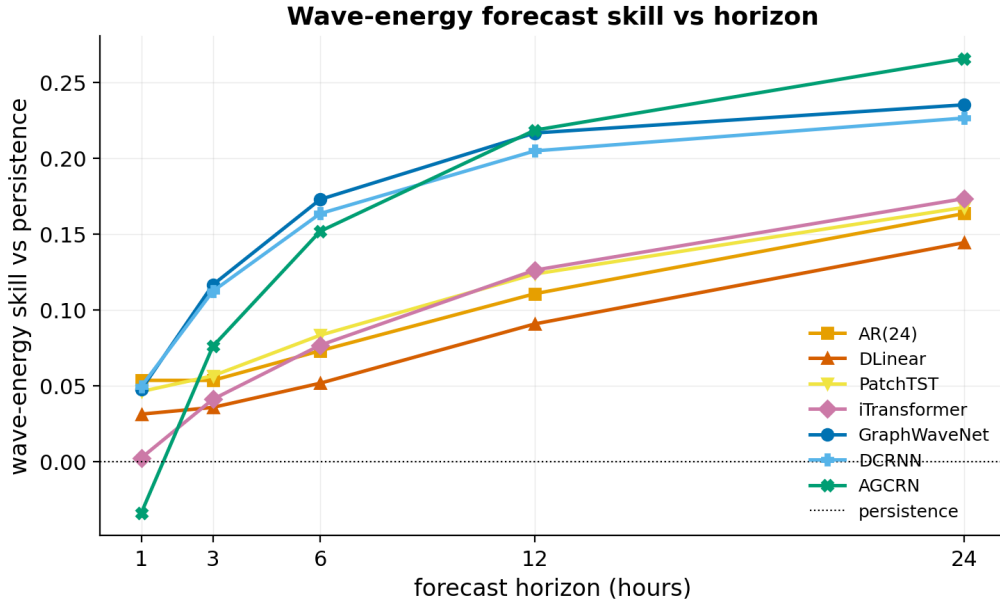


Figure 13: Wave-energy forecast skill versus horizon. The two-tier split by paradigm carries into energy and widens: the graph models pull further ahead of the temporal methods than in the height/period forecasts, because power scales with the square of wave height.